

AI for Lean Manufacturing - Course Activities

Ten progressive, case-based activities align with the registered Lean Manufacturing outcomes and the WA/Practical Performance assessment map. Each activity is self-contained and includes editable Markdown, printable PDF counterparts, evidence checklists and mock data where required.

Activity	Title	Topic	Mapping	Folder
1	Waste Walk and AI Opportunity Scan	1	K1 K2 A1	activity-01-waste-walk
2	SIPOC and Value-stream Decision Map	1	K4 A1	activity-02-sipoc-value-stream
3	Capacity, Takt and Bottleneck Analysis	1	K4 A4	activity-03-capacity-takt
4	ABC Inventory Classification with Criticality	2	K5 K6 A2	activity-04-abc-pareto
5	EBQ and Kanban Control Design	2	K7 A3	activity-05-ebq-kanban
6	Root Cause with 5 Whys and Fishbone	2	K7 A6	activity-06-root-cause
7	AI Visual Quality Pilot	3	K3 A6	activity-07-ai-quality
8	Predictive-maintenance Decision Design	3	K3 A5	activity-08-predictive-maintenance
9	Grounded GenAI for Standard Work	3	K8 A6	activity-09-genai-standard-work
10	Lean-AI Pilot Charter and Control Plan	3	K1-K8 A1-A6	activity-10-lean-ai-capstone

Interactive tools

- 5 Whys: <https://alfredang.github.io/5whys/>
- Fishbone: <https://alfredang.github.io/fishbone/>
- Pareto chart: <https://alfredang.github.io/paretochart/>
- System loop: <https://alfredang.github.io/systemloop/>
- SIPOC: <https://alfredang.github.io/sipoc/>